

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgavi-590018,Karnataka,India



A REPORT

On

“AGRO-ECOLOGY: Using ESP32 to Harmonize Wildlife and Crop Sustainability”

Submitted in Partial Fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

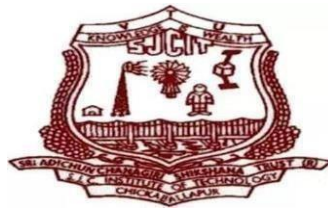
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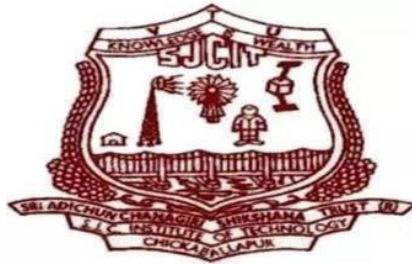
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CERTIFICATE



This is to certify that the project work entitled “AGRO-ECOLOGY: Using ESP32 to harmonize wildlife and crop sustainability” is a bonafied work carried out by **Harini M S (1SJ21EC053)**, **Monisha D(1SJ2EC084)**, **Priyanka C V (1SJ21EC112)** in partial fulfillment for the award of **Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi** during the year **2023-2024**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements with respect to project work prescribed for the Bachelor of Engineering degree.

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DECLARATION

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ABSTRACT

Crop raiding refers to the phenomenon where wild animals enter cultivated land and consume or damage crops[1]. This behavior is a significant issue in areas where wildlife habitats overlap with agricultural lands, leading to conflicts between humans and wildlife. Crop raiding can cause substantial economic losses for farmers and can sometimes result in retaliatory actions against the wildlife. Crop raiding in India significantly impacts agriculture, livelihoods, and wildlife conservation. Economic losses from destroyed crops by animals like elephants, wild boars, monkeys, and nilgai are substantial, particularly for smallholder farmers[3]. These losses threaten food security and require investments in mitigation measures, further straining financial resources. Human safety is also a concern, with injuries and fatalities resulting from encounters with large animals, causing stress and fear among rural populations. Our project is centered on safeguarding crops against animals like elephants, monkeys, and specific birds. Precautionary measures are the first thing we take to prevent animals from entering the crop[5]. The next step would be to keep them safe before informing the farmer about them. In our project, we accomplish this by activating the LEDs and utilizing a buzzer to make auditory noises that frighten the animals. Ultrasonic sensors and Force resistant sensors, which are located roughly 20-25 cm away from the field in our modest project and may range from a few kilometers in the real world, aid in the detection of the animal from a distance[8]. There is just one direction in which the ultrasonic sensor may identify impediments. We will use a force sensor to distribute it across the entire field, and when an animal treads on it, a message is sent to farmer's mobile phones, alerting them to the need to protect their farm fields. This measure is conducted when the animal is in or near farmland, allowing farmers to get to the farm before the animal invades it. As a first step toward crop protection, we might effectively prevent invaders from destroying crops.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Agriculture has always been at the heart of human civilization, but the traditional methods often lead to conflicts between crop production and wildlife conservation. The need to produce enough food for a growing population has historically led to practices that can harm the environment and biodiversity. However, with the advent of modern technology, new solutions are emerging that can help bridge this gap. This report explores the innovative use of ESP32, a powerful and versatile microcontroller, in agro-ecology to promote sustainability and harmony between wildlife and agricultural practices. By leveraging the capabilities of ESP32, farmers and researchers can develop and implement smart solutions that enhance crop yield while preserving natural habitats and supporting biodiversity.

1.2 OVERVIEW

The project "Agro-Ecology: Using ESP32 to Harmonize Wildlife and Crop Sustainability" seeks to incorporate the ESP32 microcontroller into agricultural methods in order to strike a balance between wildlife conservation and crop production sustainability. This program uses environmental sensors to evaluate soil moisture, temperature, humidity, and light intensity, as well as wildlife monitoring equipment like cameras, motion detectors, and sound sensors, to track wildlife activity near crop fields. Crop health sensors, such as the NDVI (Normalized Difference Vegetation Index) and soil nutrient sensors, are also used to evaluate crop health and growth. The ESP32 takes real-time data from various sensors, interprets it locally, and sends it to a central server or cloud platform for advanced data analytics and machine learning- driven decision-making. A user-friendly web or mobile application The interface offers farmers with real-time data, historical trends, and alarms via accessible dashboards and visualization tools, allowing them to make informed, data-driven decisions. This project's benefits include increased crop yield through optimal resource utilization, lower environmental impact, fewer human-wildlife confrontations, and overall greater efficiency and sustainability in agricultural methods. Sensor accuracy, data privacy, cost accessibility, and giving proper farmer training are all challenges that are studied and handled throughout the project

1.3 PROBLEM STATEMENT

“AGRO-ECOLOGY: using ESP32 to harmonize wildlife and crop sustainability”

The crop raiding is a significant issue in areas where wildlife habitats overlap with agricultural lands, leading to conflicts between humans and wildlife. Crop raiding in India significantly impacts agriculture, livelihoods, and wildlife conservation. Economic losses from destroyed crops by animals like elephants, wild boars, monkeys, and nilgai are substantial, particularly for smallholder farmers. Our project is centered on safeguarding crops against animals like elephants, monkeys, and specific birds. Precautionary measures are the first thing we take to prevent animals from entering the crop. The next step would be to keep them safe before informing the farmer about them.

1.4 SIGNIFICANCE AND RELEVANCE

As global food demand rises due to population growth, sustainable farming practices become imperative to ensure food security without degrading the environment. By integrating the ESP32 microcontroller and various sensors into farming systems, this project enables precise monitoring and management of environmental conditions, crop health, and wildlife activity.

Efficiency in Sustainable Agriculture:

Time Savings: Traditional farming methods can be labor-intensive and time-consuming, often requiring significant manual effort to monitor crops and wildlife. The integration of the ESP32 microcontroller automates data collection and processing, drastically reducing the time needed for these tasks.

High Volume Handling: The system is designed to handle large volumes of environmental and wildlife data efficiently, ensuring comprehensive monitoring and management without the limitations of manual methods.

Precise Monitoring: Using advanced sensors and IoT technology, the project ensures accurate measurement of soil moisture, temperature, humidity, light intensity, and other critical parameters, leading to better-informed agricultural decisions.

Contextual Understanding: The system can interpret the context of wildlife activity and crop health data, providing nuanced insights that go beyond simple metric readings.

Unbiased Decision-Making:

Reduction of Human Error: Human observations in agriculture can be prone to error and bias. The ESP32-based system evaluates all data objectively, based solely on sensor readings and analytics.

Promoting Sustainable Practices: By focusing on accurate data collection and analysis, the project supports more sustainable and environmentally friendly farming practices, reducing the reliance on subjective decision-making.

Integration with Existing Systems:

Seamless Integration: The ESP32-based solution can be integrated with existing farm management systems, enhancing their functionality without requiring a complete overhaul of current agricultural practices.

Actionable Insights: Farmers receive detailed reports and actionable insights, enabling them to make more informed decisions based on comprehensive data analysis.

Continuous Improvement:

Customization: The technology can be tailored to the specific needs of different crops and regions, ensuring relevance across various agricultural contexts.

1.5 OBJECTIVES

- To identify animals by integrating force and ultrasonic sensors.
- The next course of action would be to protect them before letting the farmer know about them. We achieve this in our project by turning on the LEDs and using a buzzer to produce loud noises that terrify the animals.
- To establish human-nature coexistence.

1.6 METHODOLOGY

The project aims to develop an effective system for protecting crops from wildlife threats, including monkeys, elephants, and various bird species. The methodology is structured to provide comprehensive protection through a multi-tiered approach involving detection, deterrence, and real-time alerts. Below is an elaboration of the proposed methodology:

1. Precautionary Measures

Objective: To establish initial deterrents and monitoring mechanisms that prevent animals from entering the crop fields.

Implementation:

Barrier Setup: Physical barriers or deterrents, such as fences or protective nets, may be installed around the crop area. These barriers act as the first line of defense to prevent animals from accessing the crops.

2. Detection and Deterrence System

Objective: To identify and deter animals before they can reach the crops, thereby minimizing potential damage.

Components and Functionality:

Ultrasonic Sensors:

Placement: Positioned around the perimeter of the crop field at intervals of 20-25 cm. In a real-world application, these sensors can be spaced further apart to cover larger areas.

Function: Ultrasonic sensors emit high-frequency sound waves that are inaudible to humans but distressing to animals. These sound waves create a protective zone around the crops, discouraging animals from approaching.

Force Sensors:

Placement: Embedded in areas of the field where animals are likely to tread. These sensors are particularly useful for detecting animals that bypass the ultrasonic barriers.

Function: Force sensors detect the pressure exerted by an animal's weight when it steps on them. This provides a more localized detection of animal presence and complements the ultrasonic sensors.

LEDs and Buzzers:

Activation: When either the ultrasonic or force sensors detect animal activity, LEDs and buzzers are activated.

Function: The LEDs emit bright flashes of light, and the buzzers produce loud acoustic sounds. These stimuli are intended to startle and repel animals, preventing them from approaching or remaining in the crop area.

3. Real-Time Alert System

Objective: To notify the farmer promptly about animal intrusions, enabling timely intervention. Components and Functionality:

Bluetooth Interface:

Integration: The Bluetooth module is connected to the force sensors.

Function: When the force sensors detect an animal, the Bluetooth interface transmits a notification to the farmer's mobile phone. This alert includes details such as the time and location of the intrusion, allowing the farmer to take immediate action.

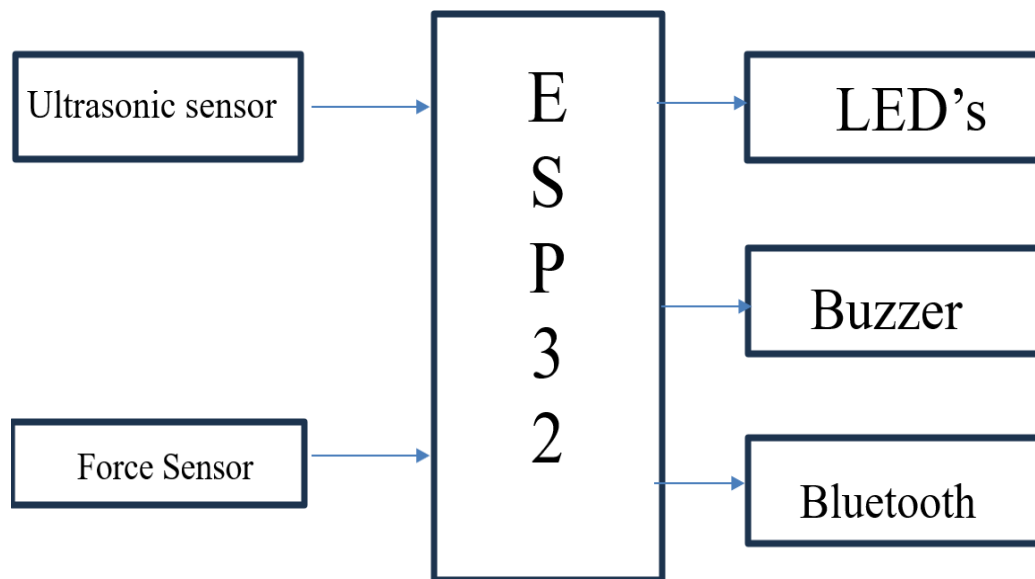


Fig 1.6.1 Block Diagram

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

Catherine M. Hill et al. [1] This paper has proposed the study, which uses systematic monitoring of the behavior of non-human primates on farms in western Uganda. The number of individuals raiding and the duration of the raid are the major characteristics determining crop loss. Secondary considerations include the distance traveled to the farm, the age mix of the raiding group, and whether the raids are in sequence.

D Naha et al. [2] Examined a 2-year dataset of agricultural depredation occurrences by Asian elephants (N = 380) in North Bengal (eastern India). We also investigated the impact of landscape and anthropogenic factors (area of forest, farmland, distance from protected area, area of human settlements, riverine patches, and population density) on the spatial occurrence of such instances. The results of our spatial analysis suggest that crop depredation rose with an increase in the area of forest patches, agriculture, riverine patches, and human density.

Vijay Kumar et al. [3], This paper has proposed that the primary triggers of crop raiding were a loss of grazing places due to deforestation, a decline in predatory animals, lantana infestation in the forests, and overexploitation of the forests for wild fruits. Farmers traditional approaches to these concerns were inadequate against the vast majority of these species. Some methods for lowering crop raiding losses include categorizing some animals as vermin, sterilizing monkeys, paying injured farmers, carefully enforcing wildlife laws, and eradicating lantana.

Meenakshi Negi et al [4], This study was undertaken to evaluate factors which potentially influence crop depredation by elephants during the non- cropping season. He proposed that crop raiding can be controlled by practicing diversification of crops by shifting to pulses and some major efforts might be done for the marketing of those pulses and local agricultural products. Solar electric fencing can be a better option to mitigate the problem.

Robin Rathi [5], This finding is consistent with results from studies of primate foraging in risky environments, where animals remain near the edge of high-risk habitats (Cowlshaw 1997), and typically are in larger groups under higher-risk conditions (Hill & Lee 1998). However, studies of crop- foraging baboons in Nigeria (Warren 2003), Uganda (Wallace & Hill 2012), and Zimbabwe (Schweitzer et al. 2017) and elephants in Botswana (Songhurst 2017)

Latika Pandey et al. [6], The current research is a case study of this issue in Malai Mahadeshwar Wildlife Sanctuary (MM Hills) of Western Ghats in South India, with a special emphasis on elephant depredation. The objective of the study is to understand the types of humans–wildlife conflict in selected villages.

Riju P Nair et al [7], Crop raiding is a major form of human–wildlife conflict that not only affects livelihoods of farmers living close to forest areas but also jeopardizes the objective of wildlife conservation. In this study, we report patterns associated with crop raiding based on periodic field inspections of 95 crop fields spread across 16 villages in India.

Milind D Patil et al [8], Range extension of elephants has been observed in northern Western Ghats over the past few decades. The Sindhudurg District is one such area experiencing recent colonization by elephants. We obtained data on crop raiding incidences between 2002 and 2015 from Range Forest Offices in Sindhudurg District. Information was available on 9148 crop raiding cases.

Suresh K Govind et al [9], This study was undertaken to evaluate factors which potentially influence crop depredation by Asian elephants during the non-cropping season. The study was conducted in a 2.5 km zone abutting Bannerghatta Wildlife Range of Bannerghatta National Park, Karnataka. Three fruit trees which are commonly raided by elephants, namely jackfruit (*Artocarpus heterophyllus*), mango (*Mangifera indica*) and tamarind (*Tamarindus indica*) were selected for the study.

Leta Goboshoa et al [10], The raiding of crops by elephants is one of the major components of human-elephant conflict, causing loss of livelihood and retaliation against elephants. To mitigate this conflict, various intervention methods are in use by farmers across Africa and Asia; yet there have been few rigorous assessments of their effectiveness.

Catherine M. Hill et al [11], This finding is consistent with results from studies of primate foraging in risky environments, where animals remain near the edge of high-risk habitats (Cowlishaw 1997), and typically are in larger groups under higher-risk conditions (Hill & Lee1998).

CHAPTER 3

SYSTEM REQUIREMENTS AND SPECIFICATION

3.1 HARDWARE SPECIFICATION

The project utilizes ESP32 microcontroller along with sensors like Ultrasonic sensor, Force sensors, LED's, Buzzers and ESP Cam as an extension to help us in achieving the set target.

ESP32: Wireless Communication, Digital Interfaces, Processing Power, Real-time Operating System, Analog Inputs, Digital Interfaces are few functions of ESP32, we have also adopted these functions in our project, we make use of ESP32 for even Bluetooth interfacing.

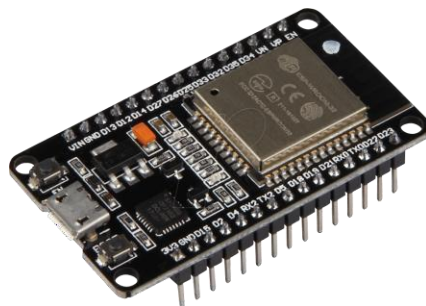


Fig 3.1.1 ESP32 Controller

Ultrasonic sensor: The ultrasonic sensor is a non-contact type of sensor used to measure an object's distance and velocity. This sensor operates on sound wave property to measure the velocity and distance of the object



Fig 3.1.2 Ultrasonic Sensor

Force sensor: A force sensor, which is also known as a load cell or force transducer, is a sensor that measures force by converting the input of mechanical force into the output of an electrical signal. As the force is applied to the sensor, its electrical output signal can be measured, converted, and standardized



Fig 3.1.3 Force Sensor

LED's: Display Technology, Indicators, and Communications, make use of LED's. In our project we made use of LED's, to indicate fire which may create fear in the rodents.

Buzzer: Buzzer meaning electronic component that generates sound through the transmission of electrical signals. Its primary function is to provide an audible alert or notification and typically operates within a voltage range of 5V to 12V.

3.1 SOFTWARE REQUIREMENT

Arduino ide: The Arduino IDE (Integrated Development Environment) is a software application used for writing, compiling, and uploading code to Arduino microcontroller boards.



Fig 3.1.4 Arduino IDE

3.2 FUNCTIONAL REQUIREMENT

Animal Detection System: The system shall detect animals approaching the farm land using ultrasonic sensors and pressure sensors.

Alert Mechanism: The system shall activate visual (LEDs) and auditory (buzzers) deterrents when animals are detected.

Farmer Notification: The system shall send a notification to the farmer's mobile phone when an animal is detected near or within the farmland.

Sensor Network Coverage: The system shall provide scalable sensor coverage that can be extended to cover several kilometers of farmland.

Power Management: The system shall be capable of operating continuously in a farmland environment, with considerations for low power consumption.

Environmental Resilience: The system shall be designed to withstand outdoor environmental conditions, such as rain, wind, and temperature fluctuations.

False Alarm Minimization: The system shall include logic to minimize false alarms caused by non-threatening objects or environmental factors.

These requirements outline the basic functionality needed for your crop protection system to be effective in real-world applications.

3.3 NON-FUNCTIONAL REQUIREMENTS

Here are some non-functional requirements for your crop protection project:

Performance: The system should detect animals and activate deterrents within 1-2 seconds of detection.

Reliability: The system should operate with a detection accuracy of at least 95%.

Scalability: The system should be able to scale from small to large farms without significant changes to the architecture.

Maintainability: The system should be easy to maintain and update with minimal technical expertise.

Usability: The system's interface for notifications should be simple and intuitive for farmers to understand and respond to alerts.

Power Efficiency: The system should consume low power and be able to operate on batteries or solar power for at least 6 months without maintenance.

CHAPTER 4

SYSTEM ANALYSIS

4.1 EXISTING SYSTEM

Similar to this there are existing systems and technologies designed to protect crops from animal invasions. These systems typically use a combination of sensors, deterrents, and communication technologies similar to what your project is developing. Here are a few examples:

Electronic Scarecrows:

Functionality: These systems use motion sensors or infrared sensors to detect approaching animals and then activate deterrents like loud noises, flashing lights, or water sprays to scare them away.

Acoustic Deterrent Systems:

Functionality: These systems emit high-frequency sounds or distress calls of specific animals to deter them. Ultrasonic repellers are common for small animals and birds.

Fence Systems with Sensors:

Functionality: Electric fences integrated with sensors can detect large animals like elephants. When an animal is detected, the fence can deliver a mild electric shock, or other deterrent mechanisms like alarms or lights are activated.

Drones for Wildlife Monitoring:

Functionality: Drones equipped with cameras and sensors are used to monitor large areas of farmland. They can detect animal movement and send real-time alerts to farmers.

Smart Farm Management Systems:

Functionality: These systems integrate various IoT devices, including sensors for animal detection, and provide real-time data and alerts to farmers via mobile apps.

4.1.1 Drawbacks

Electronic Scarecrows:

Habituation: Animals may become accustomed to the deterrents over time, reducing their effectiveness.

Limited Range: These systems typically cover a small area, making them less effective for larger farmlands.

Acoustic Deterrent Systems:

Animal Habituation: Animals may learn to ignore the sounds if exposed frequently, decreasing the system's long-term effectiveness.

Non-target Species Impact: High-frequency sounds may affect non-target animals or even domesticated animals nearby.

Fence Systems with Sensors:

High Cost: Installing and maintaining electric fences, especially over large areas, can be expensive.

Maintenance: Regular maintenance is needed to ensure the fence remains functional, as animals or environmental factors can damage it.

Drones for Wildlife Monitoring:

High Initial Cost: Drones and associated technology can be expensive to acquire and operate.

Short Battery Life: Drones have limited flight time due to battery constraints, reducing their monitoring capabilities.

Smart Farm Management Systems:

Complexity: These systems can be complex to install and configure, especially for small-scale or low-tech farmers.

High Cost: The initial cost of these systems, including sensors and integration with other farm management tools, can be prohibitive.

4.2 PROPOSED SYSTEM ADVANTAGES

Here are the advantages of our crop protection project:

Proactive Animal Detection:

Advantage: By using ultrasonic and pressure sensors, the system detects animals early, before they reach the crops. This proactive approach helps in preventing damage rather than just reacting to it.

Impact: Reduces crop loss and ensures timely intervention, protecting the farmer's livelihood.

Multi-Sensory Detection:

Advantage: The combination of ultrasonic sensors for distance detection and force sensors for pressure detection provides comprehensive coverage, ensuring that animals approaching from different directions and at various distances are detected.

Impact: Increases the reliability and accuracy of the detection system, minimizing the chances of animals slipping through undetected.

Immediate Deterrent Mechanism:

Advantage: The activation of LEDs and buzzers serves as an immediate deterrent, scaring away animals before they can cause harm to the crops.

Impact: Reduces the need for the farmer to be present at all times, as the system can autonomously deter animals.

Real-Time Notifications:

Advantage: The system sends real-time alerts to the farmer's mobile phone when animals are detected near or within the farmland.

Impact: Enables the farmer to take prompt action if needed, enhancing the overall effectiveness of the crop protection system.

Scalability:

Advantage: The system is designed to be scalable, with sensors placed 20-25 cm apart in the mini-project setup, but it can be expanded to cover several kilometers in a real-world application.

Impact: Adaptable for different farm sizes, from small to large-scale operations, making it a flexible solution.

Cost-Effectiveness:

Advantage: By using commonly available sensors and simple electronic components like LEDs and buzzers, the system can be implemented at a relatively low cost.

Impact: Makes the solution accessible to smallholder farmers who may not have the resources to invest in more expensive technologies.

Low Maintenance:

Advantage: The system's components are relatively simple and durable, requiring minimal maintenance compared to more complex systems like drones or camera-based detection.

Impact: Reduces long-term operational costs and ensures continuous protection with less frequent maintenance.

Energy Efficiency:

Advantage: The system can be designed to be energy-efficient, with low power consumption for sensors, LEDs, and buzzers, potentially operating on batteries or solar power.

Impact: Suitable for deployment in remote areas without reliable electricity, ensuring consistent operation.

Environmentally Friendly:

Advantage: The use of non-lethal deterrents like lights and sounds is environmentally friendly, posing no harm to the animals or the surrounding ecosystem.

Impact: Supports sustainable farming practices by preventing animal intrusion without causing ecological damage.

Minimization of False Alarms:

Advantage: The dual-sensor approach (ultrasonic and force sensors) helps in reducing false positives, ensuring that alerts are sent only when a genuine threat is detected.

Impact: Enhances the trustworthiness of the system, preventing unnecessary alerts and ensuring that the farmer only responds when necessary.

CHAPTER 5

SYSTEM DESIGN

5.1 PROJECT MODULES

1. Sensor Integration Module

To integrate and configure the ultrasonic and pressure (force) sensors for animal detection.

Components:

Ultrasonic Sensors: Detect animals from a distance.

Pressure Sensors: Detect the presence of animals when they step on the sensor.

Sensor Calibration: Ensure that the sensors are accurately calibrated for distance and sensitivity to minimize false positives.

Key Tasks:

Interfacing sensors with the microcontroller. Developing algorithms for processing sensor data. Testing and calibrating sensors for optimal performance.

2. Detection and Deterrent Activation Module

To process sensor data and activate deterrent mechanisms (LEDs and buzzers) when an animal is detected.

Components:

Microcontroller: Processes input from sensors and controls the deterrent devices. LEDs

and Buzzers: Act as visual and auditory deterrents to scare away animals.

Key Tasks:

Programming the microcontroller to trigger LEDs and buzzers based on sensor inputs.

Configuring deterrent mechanisms to operate effectively without causing harm.

Testing the activation timing and deterrent effectiveness in various scenarios.

3. Communication and Notification Module

To send real-time alerts to the farmer's mobile phone when an animal is detected.

Mobile Application or Web Dashboard: Interface where the farmer receives alerts and monitors the system status.

Key Tasks:

Developing the notification system using GSM or an IoT platform. Designing a simple mobile application or web interface for alerts.

Testing the reliability and speed of notifications required (eg. battery replacement)

5.2 DATAFLOW DIAGRAM

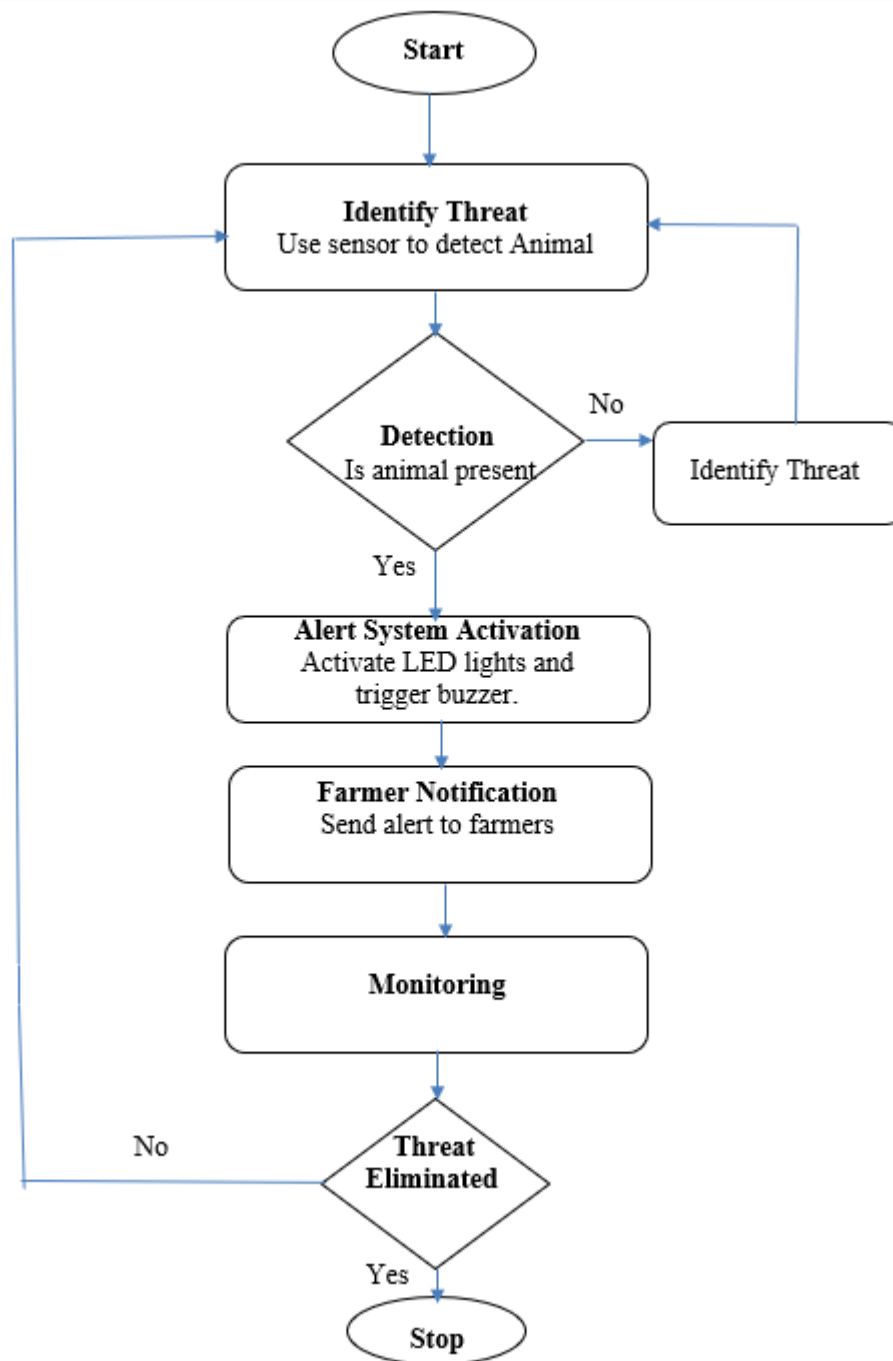


Fig 5.2.1 Flowchart

CHAPTER 6

IMPLEMENTATION

6.1 ALGORITHM

Initialization:

1. Initialize sensors:
 - Set up the ultrasonic sensors.
 - Set up the pressure sensors.
 - Set up LEDs and buzzers.
 - Initialize communication module (GSM/IoT) for sending alerts.
2. Set detection parameters:
 - Define the detection threshold for ultrasonic sensors (e.g., distance to detect an approaching animal).
 - Define the pressure threshold for force sensors.

Main Loop:

1. Monitor Ultrasonic Sensors:
 - Continuously monitor ultrasonic sensors for detecting any approaching animal.
 - If an object (animal) is detected within the defined threshold distance, proceed to the next step.
2. Activate Deterrents:
 - Trigger LEDs to flash.
 - Trigger buzzers to emit sound.
 - Wait for a short period (e.g., 10-20 seconds) to see if the animal moves away.
3. Check for Animal Proximity:
 - After activating deterrents, monitor pressure sensors for any steps on the sensors.
 - If pressure is detected, the animal is still within or near the farmland.
4. Send Alert to Farmer:
 - If pressure is detected or if ultrasonic sensor continuously detects the animal after deterrents have been activated, send an alert to the farmer's mobile phone.
 - Include information like the location of detection, time, and possible type of animal.
5. Recheck Situation:
 - After sending the alert, continue monitoring ultrasonic and pressure sensors.
 - If the animal leaves the detection area, reset the system to initial state.
6. Loop Continuation:
 - Continue the loop for ongoing monitoring and protection.

End:

- The system runs continuously to monitor and protect the crops.

6.2 CODE:

```
#include <BluetoothSerial.h>

// Define pins for ultrasonic sensors
const int trigPin1 = 2;
const int echoPin1 = 4;
const int trigPin2 = 18;
const int echoPin2 = 5;
const int trigPin3 = 21;
const int echoPin3 = 19;

// Define pin for LDR
const int ldrPin = 32;

// Define pins for LEDs
const int ledPin1 = 25;
const int ledPin2 = 26;
const int ledPin3 = 27;
const int ledPin4 = 14;

// Define pin for buzzer
const int buzzerPin = 15;

// Bluetooth serial
BluetoothSerial SerialBT;

void setup() {
    // Initialize serial communication
    Serial.begin(115200);
    SerialBT.begin("ESP32_Animal_Detector");

    // Initialize pin modes
    pinMode(trigPin1, OUTPUT);
    pinMode(echoPin1, INPUT);
    pinMode(trigPin2, OUTPUT);
    pinMode(echoPin2, INPUT);
    pinMode(trigPin3, OUTPUT);
    pinMode(echoPin3, INPUT);
    pinMode(ldrPin, INPUT);
    pinMode(ledPin1, OUTPUT);
    pinMode(ledPin2, OUTPUT);
    pinMode(ledPin3, OUTPUT);
```

```
pinMode(ledPin4, OUTPUT); pinMode(buzzerPin, OUTPUT);
}

void loop() {
  bool animalDetected = false;
  int distance1 = checkUltrasonic(trigPin1, echoPin1);

  int distance2 = checkUltrasonic(trigPin2, echoPin2);

  int distance3 = checkUltrasonic(trigPin3, echoPin3);

  if (distance1 <= 100 && distance1 > 0)
  {
    animalDetected = true;
    SerialBT.printf("Animal detected by sensor 1 at %d cm\n", distance1);
  }
  if (distance2 <= 100 && distance2 > 0)
  {
    animalDetected = true;
    SerialBT.printf("Animal detected by sensor 2 at %d cm\n", distance2);
  }
  if (distance3 <= 100 && distance3 > 0)
  {
    animalDetected = true;
    SerialBT.printf("Animal detected by sensor 3 at %d cm\n", distance3);
  }

  // Check LDR
  if (analogRead(ldrPin) > 500)
  {
    animalDetected = true;
    SerialBT.println("Animal detected in the vicinity of the forest!");
  }
}
```

```
if (animalDetected)
{
digitalWrite(ledPin1, HIGH);
digitalWrite(ledPin2, HIGH);
digitalWrite(ledPin3, HIGH);
digitalWrite(ledPin4, HIGH);
activateBuzzer();
}
Else
{
digitalWrite(ledPin1, LOW);

digitalWrite(ledPin2, LOW);
digitalWrite(ledPin3, LOW);
digitalWrite(ledPin4, LOW);
digitalWrite(buzzerPin, LOW);
}
delay(500); // Small delay to avoid excessive checks
}

int checkUltrasonic(int trigPin, int echoPin) {long duration;

int distance;

digitalWrite(trigPin, LOW); delayMicroseconds(2); digitalWrite(trigPin, HIGH);
delayMicroseconds(10); digitalWrite(trigPin, LOW);

duration = pulseIn(echoPin, HIGH);distance = duration * 0.034 / 2;
if (distance <= 100) { // Check if the distance is within 100 cmreturn distance;
} else { return -1;
}
}

void activateBuzzer() {
for (int i = 0; i < 3; i++)

{
delay(500); noTone(buzzerPin);delay(500);
}
}
```

CHAPTER 7
RESULTS AND DISCUSSIONS

7.1 RESULTS

The project focuses on protecting crops from animals such as monkeys, elephants, and certain birds using a system that integrates ultrasonic sensors, pressure sensors, LEDs, buzzers, and a Bluetooth communication module. The system is designed to detect the presence of animals and deter them before sending a notification to the farmer. The steps involved in setting up the system, assembling the components, and the results observed are outlined below.

Step 1: Connection of Microcontroller with Sensors and Deterrent Devices

As shown in the corresponding diagram, the first step involves connecting various sensors and deterrent devices to a microcontroller (e.g., Arduino, Raspberry Pi).

Ultrasonic Sensors:

VCC: Connected to the 5V pin of the microcontroller to provide power. GND: Connected to the ground pin of the microcontroller.

Trigger Pin: Connected to a digital I/O pin (e.g., D2) to send ultrasonic waves.

Echo Pin: Connected to another digital I/O pin (e.g., D3) to receive reflected waves and calculate the distance.

Pressure Sensors (Force Sensors):

VCC: Connected to the 3.3V or 5V pin, depending on the sensor. GND: Connected to the ground pin of the microcontroller.

Output Pin: Connected to an analog input pin (e.g., A0) to detect pressure applied by animals stepping on it.

LEDs and Buzzers:

LEDs: The anode is connected to a digital I/O pin (e.g., D4) through a resistor, and the cathode is connected to GND.

Buzzer: The positive terminal is connected to a digital I/O pin (e.g., D5).

Step 2: Connecting the Microcontroller to a Bluetooth

Turn on the Bluetooth in your mobile phone and open serial Bluetooth app and connected to the esp32 controller board as specified in the code.

After connecting the components, they were assembled into a working model as illustrated in the corresponding setup diagram. This setup was tested in a controlled environment to simulate real-world conditions.

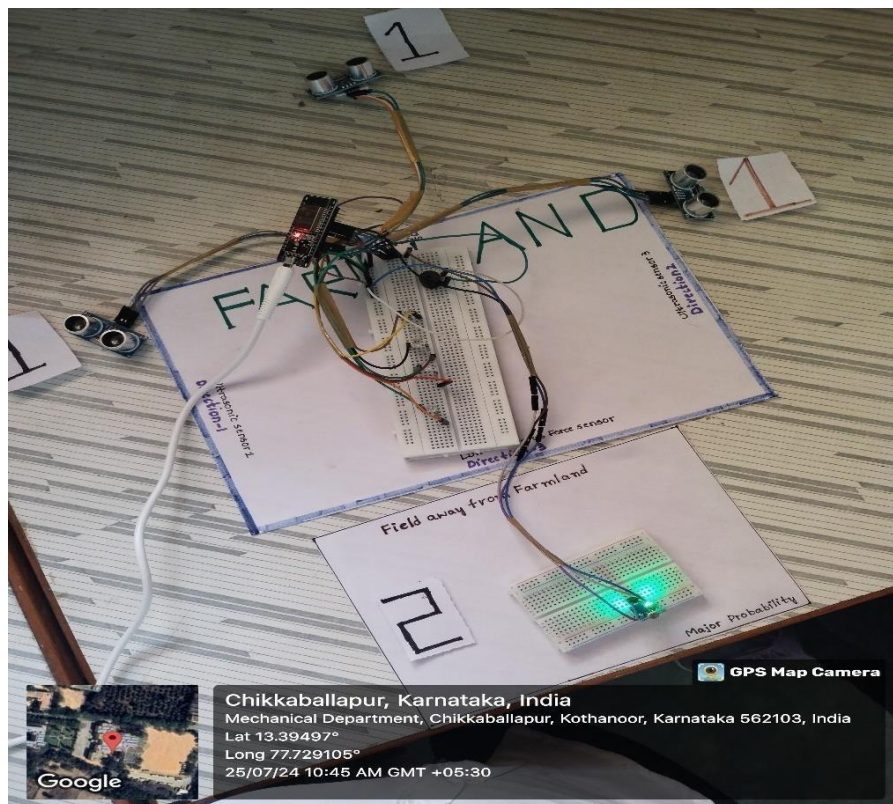


Fig.7.1.1 Experimental setup

Testing of Ultrasonic Sensors:

The ultrasonic sensors successfully detected obstacles (representing animals) at varying distances, triggering the deterrent mechanisms (LEDs and buzzers) appropriately.

Testing of Force Sensors:

The force sensors accurately detected the presence of animals stepping on them, demonstrating the ability to detect ground-based intrusions.

Deterrent Activation:

When an animal was detected by either sensor, the LEDs flashed and the buzzer emitted a sound, effectively simulating a deterrent response to scare the animals away.

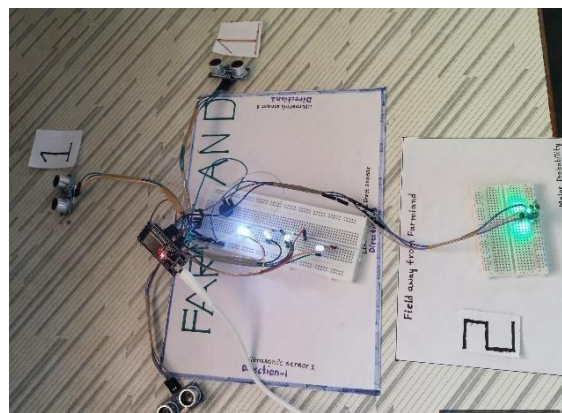


Figure 7.1.2 Activation of LED's and Buzzer

Notification System:

Upon detection, the system sent alerts via Bluetooth to the farmer's mobile phone using a Bluetooth terminal application. The notifications were successfully tested by triggering the sensors and verifying the receipt of alerts on the connected device.

The integrated system effectively detected the presence of animals and responded with deterrent actions before sending alerts to the farmer. The use of both ultrasonic and pressure sensors provided a robust detection mechanism for different types of intrusions.

The deterrent mechanism, consisting of visual (LEDs) and auditory (buzzers) signals, was effective in simulating the scaring away of animals.

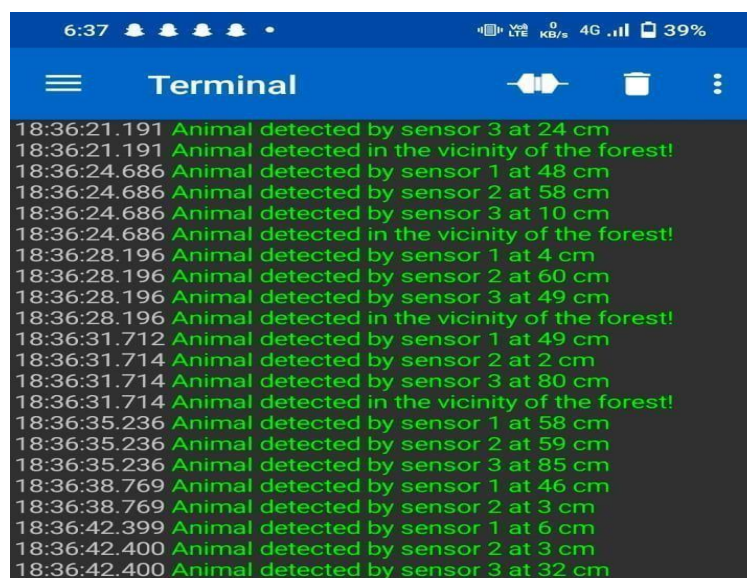


Figure 7.1.3 Notification System

7.2 DISCUSSIONS

System Effectiveness:

The combination of ultrasonic and pressure sensors enhanced the system's ability to detect animals from different approaches. The deterrent actions were promptly triggered, reducing the likelihood of animals reaching the crops.

Challenges:

Ensuring proper sensor placement and calibration was critical for accurate detection. Environmental factors like weather conditions or sensor alignment could potentially affect performance, and these aspects were carefully considered during testing.

Scalability:

The system demonstrated potential for scalability, with the ability to cover larger areas by adding more sensors and extending the communication range. The modular design allows for easy expansion to suit different farm sizes.

Future Enhancements:

Potential improvements include integrating advanced detection algorithms, using solar power for remote operations, and enhancing the mobile application interface for better farmer interaction.

CHAPER 8

APPLICATIONS

8.1 APPLICATIONS

Wildlife Preservation: Non-lethal methods ensure animals are deterred without harm.

Crop Protection: Prevents wildlife from damaging crops, increasing yield and reducing economic losses.

Cost-Effective: Using affordable and readily available components like ESP32

Versatile: With built-in Wi-Fi and Bluetooth, ESP32 is suitable for a wide range of applications and can easily connect to various sensors and devices.

Low Power Consumption: Ideal for remote locations where power resources may be limited.

Open-Source: Being an open-source platform, it allows for customization and community-driven improvements.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENT

9.1 CONCLUSION

The crop raiding project aimed to assess the impact of wildlife on agricultural productivity and develop measures to mitigate such conflicts. Our comprehensive investigation revealed substantial economic losses for farmers due to crop damage inflicted by various wildlife species. Data collected from field observations, farmer interviews, and remote sensing technology highlighted the patterns and frequency of crop raids, identifying the primary animal culprits and the most damaged crop varieties.

The findings underscore the necessity of integrating wildlife management with agricultural practices. Effective mitigation strategies, such as physical barriers, community-based monitoring systems, and non-lethal deterrents, proved essential in reducing crop raiding incidents. Additionally, promoting wildlife-friendly farming techniques and fostering collaboration between farmers and conservationists can help mitigate conflicts. Crop raiding incidents were more common in fields near forest edges, with certain crops like maize and bananas being frequently targeted.

9.2 FUTURE ENHANCEMENT:

Integration with Advanced AI and Machine Learning:

Animal Identification: The system can be enhanced with AI and machine learning algorithms to identify different types of animals based on their movement patterns, size, and behavior. This would allow for customized deterrent strategies based on the specific animal detected.

Predictive Analytics: Implementing predictive analytics to anticipate potential animal invasions based on historical data, weather patterns, and time of year, enabling preemptive measures to be taken.

Solar-Powered and Energy-Efficient Design:

Renewable Energy Sources: The system can be powered by solar panels, making it suitable for remote or off-grid locations. This would also reduce operational costs and make the system more sustainable.

Low-Power Sensors and Modules: Optimizing the system with low-power sensors and communication modules to extend battery life and reduce energy consumption.

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